

AI-Enhanced Image Processing for Target Detection and Threat Recognition in Defence Applications

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Abstract— The importance of rapid and precise threat recognition has become critical in modern defence operations. In this research, we describe an AI-based image processing framework optimized for real-time detection and recognition of complex military threats. The system integrates satellite, UAV, thermal, and night vision images into a single processing pipeline which performs noise reduction, contrast enhancement as well as normalization. The context focus approach entails extraction of interest regions using motion capture techniques. For object detection and classification, the framework employs advanced deep learning models such as YOLOv8 and Faster R-CNN, with powerful feature extractors ResNet and EfficientNet. Custom CNNs process infrared images for drone, vehicle, weapon and personnel detection surmounted by low visibility conditions. Threat behaviour recognition making use of hybrid CNN-LSTM models utilizes spatial and temporospatial motion interplay. Pose estimation aids detecting aggression or concealed weaponry. Terrain comprehension deepens alongside camo differentiation through semantic segmentation employing DEeplabV3+ and U-Net. The use of Kalman Filters in multi-sensor fusion reduces the rate of false positives while improving tracking accuracy. Our system offers a reliable solution for AI-powered surveillance in defence contexts, undergoing evaluation by measuring operational performance through defined KPIs such as Precision, Recall, IoU, and FPS.

Keywords— CNN, LSTM Hybrid Models, YOLOv8, LiDAR, Infrared Image analysis, Sensor Fusion.

I. INTRODUCTION

In the cutting-edge operations of modern warfare, and national security operations in real-time, keeping track of any assessment along with danger signals has become crucial. Inaccurate threat verification provided by traditional monitoring aids due to the soaking manual interpretation and the lag due to environmental changes poses slow response timing is difficult to cater to [1]. With the advances made in AI, including computer vision and deep learning, there have been massive changes in monitoring system responsiveness and perception to scenario changes on the battlefield. Intelligent image enhancement aids in reliably countering every target or

threat aimed defining detection, registration, and deep-level technical analysis targeting new and increasingly complex scenarios without limits unparalleled speed and precision [2]. Intelligent perception systems which work reliably in highly divergent terrains and under extreme conditions, like dim light, thermal environments, or cluttered visual spaces, are essential to fulfil military applications. These systems go from UAV (Unmanned Aerial Vehicles) border surveillance to AI monitoring systems deploying deep within urban battlefields [3]. The combination of imaging deep learning with surveillance deep learning not only automates visual watchlists but assists predictive modelling and military decision making, alleviating the exhaustive reliance on manual analysis [4]. This paper presents a framework for target identification and threat detection specifically designed for military surveillance systems. This encompasses the use of convolutional neural networks (CNNs) with object detection frameworks like YOLOv8, semantic segmentation via DeepLabV3+, and activity recognition through LSTMs. Enhanced YOLO-based algorithm offering target-accomplishing detection improvements, focusing on boosting accuracy and system durability in more challenging environments, particularly for real-time object identification activities are offered [5].

A. Existing systems and their limitations

The existing defence surveillance systems employ a mixture of optical cameras, thermal cameras, radars, and satellites to monitor the regions of interest, maintaining situational awareness. Some systems use background subtraction and matched filtering followed by CNN classification. Any error made within these prior steps can severely impact detection efficiency and effectiveness. They have challenges in detection and/or target recognition beyond a certain distance due to the constrictive spatial context of small windows. [6]. Over the years, military applications have incorporated various image processing techniques, including edge detection and motion tracking, as well as template matching. While YOLOv5 is lightweight, applying it for real-time evaluation on low-

cost, resource-limited devices poses a challenge due to the still high processing needs. [7]. Automatic feature extraction and end-to-end learning from raw image data became possible due to deep learning models [8]. Now, image classification and object detection are commonly performed by Convolutional Neural Networks (CNNs) like AlexNet, VGGNet, and ResNet. In the context of military surveillance, models such as Faster R-CNN, SSD (Single Shot MultiBox Detector), and YOLO (You Only Look Once) are used for threat detection and localization of intruders, vehicles, and UAVs. The major challenge is that most available systems are tailored for general-purpose datasets, severely underperforming in the context of more complex military scenarios [9].

Difficult environments, real time interpretation, and decision-making pose great challenges to the automation provided by modern systems. Any environmental factors such as fog, smoke, nighttime, or thermal distortion make traditional image processing techniques, which rely on morphological operations and thresholding, ineffective [10]. Basic motion detection tends to produce too many false alerts from minor changes in the background due to noise, and in combination with very low thresholds, becomes unusable. In addition, traditional algorithms perform poorly on classifying objects as friend or foe, and interpreting military movements such as formations, manoeuvres, and weapon actions. They have no spatial or temporal logic for recognizing pre-defined threat patterns in sequences of activities [11-13]. The current frameworks also lack sensor data fusion, which considerably enhances the ability to construct threat maps, making them unsuitable for the defence domain.

B. Motivation for the Proposed Work

Considering these gaps, there's an urgent need to create an intelligent, accurate and adaptive framework for image processing shaped by the needs of defence applications. This research aims to change the passive surveillance system into an active threat detection system using advanced artificial intelligence and deep learning algorithms. The proposed framework is built upon several innovations, including real-time object recognition with YOLOv8, temporal pattern recognition with LSTM, and scene understanding through semantic segmentation, which improve the performance of the system [14]. The system is capable of identifying suspicious activities, detecting relevant threats such as unauthorized vehicles or personnel, and classifying behaviours in real-time by integrating spatial and temporal features. For additional situational awareness, cameras may be supplemented with LiDAR, RADAR, and infrared sensors to provide multi-sensor fusion, thus enabling full 360-degree coverage. The implementation of hybrid deep learning models improves the system's performance in detecting targets without increasing false positives [15]. More importantly, these models allow for operation in ever-changing and unpredictable conditions in-field.

The paper is structured as in Section 2, the proposed methodology is detailed. In Section 3, the description of experimental setup is provided. Section 4 is

devoted to discussion of the results. Section 5 concludes the study showcasing real-world defence systems for future development.

II. PROPOSED METHODOLOGY

The proposed methodology describes a complete and smart strategy for base image processing that facilitates real-time target and threat identification in defence settings. This system addresses shortcomings of the traditional surveillance systems employing AI techniques such as deep learning models and complex sensor fusion. The methodology can be broken down into a number of sequential steps from data collection through preprocessing, ROI selection, object identification, threat detection, scene interpretation, sensor fusion, and alert creation as shown in the fig. 1. The design criteria for these phases are robustness, precision, and flexibility to everchanging defensive conditions.

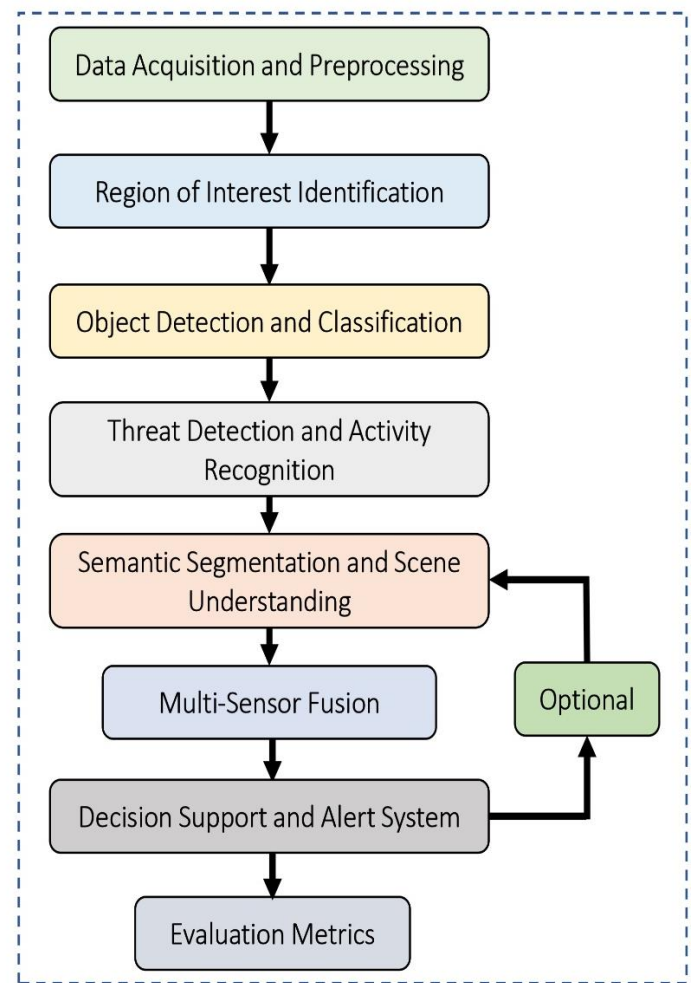


Fig. 1. Block diagram of proposed methodology

A. Data Acquisition and Preprocessing

As all image processing workflows start with the contextually rich data requirement, for defence scenarios, the range of available data is important. The different types of data which is collected includes high resolution satellite images, aerial view drone footage, videos captured on the

ground with surveillance cameras, and thermal/infrared images taken using night vision technology. This permits operations across different terrains, lighting levels, and weather. Preprocessing starts by cleaning the raw data from noise and quality issues. For denoising, Gaussian filtering and Non-Local Means (NLM) algorithms are used, where spatial noise smoothing is performed using Gaussian filters and edge structures are preserved by NLM. After the noise is addressed, imagery is further enhanced using contrast stretching and histogram equalization. In the case of video data, frame extraction is done so that only relevant sequences are captured for further analysis. Consistency is critical for optimizing the performance of deep learning models; therefore, all input data undergoes standardization for resolution, aspect ratio, and colour depth.

B. Region of Interest (ROI) Identification

As a result of the high dimensionality of surveillance imagery, analysing entire frames is rarely efficient from a computational perspective. Thus, the next step involves the locating of zones that have a high probability of containing targets or activities related to threats. This is done with the help of motion detection algorithms like the background subtraction and optical flow estimation. Background subtraction is very suitable for static surveillance configurations which derive changes over time as movement. Adaptive background modelling also accounts for lights changes or jittering of the camera. In more dynamic scenes, optical flow techniques tend to capture continuity as well as gradual shifts of various object's positions in different frames. Geofencing is employed to put limits around spatial context. Military zones and points of Interest are defined with the help of GPS coordinates or satellite overlays which aid the system in determining which ROIs to prioritize within critical zones.

C. Object Detection and Classification

Following the extraction of ROIs, the system proceeds to execute object detection and classification of different entities including personnel, military vehicles, drones and weapon systems. For this purpose, machine learning, and more specifically deep learning, paradigms are utilized. The primary model is YOLOv8 (You Only Look Once version 8), which performs single-shot detection and therefore achieves real-time performance and high accuracy simultaneously. The Capability of YOLOv8 includes detecting several classes at the same time, and it has been also proven that it can achieve fast inference with lightweight backbones such as EfficientNet or CSP Darknet. The model is first trained on large-scale datasets like MS COCO or ImageNet and subsequently fine-tuned on domain-specific military datasets containing annotated infrared, thermal, and visible spectrum images. In parallel, applications that need greater accuracy at object localization, particularly for tightly grouped or hidden objects, deploy the Faster R-CNN (Region-based Convolutional Neural Network) technique. Strategically altered convolutional neural networks are utilized to analyse thermal images where standard RGB channels are substituted with pseudo-coloured IR data to accommodate alternate spectral properties.

D. Threat Detection and Activity Recognition

Understanding behavioural patterns and intent goes beyond simply identifying objects within military contexts. In order to achieve this goal, the system incorporates hybrid deep learning models which utilize Convolutional Neural networks (CNNs) for spatial feature extraction and Recurrent Neural Networks (RNNs) like LSTM or GRU networks for capturing temporal patterns. The model is designed to flag perimeter breaches, aggressive movements, rapid troop formations, and other actions as potentially hostile. For instance, sequences that feature an individual approaching a restricted zone while brandishing a firearm will be considered threat events. Temporal models of these actions are built using feature vectors from the system's CNN backbone, allowing these models to monitor time-invariant dependencies where complex event chains, rather than actions, are the focus. Along with LSTMs, the system uses Pose Estimation Methods like Open Pose to consider human posture. From pose key points, a skeletal structure can be extracted for weapon holding and aiming, and combat positions.

E. Semantic Segmentation and Scene Understanding

Any intelligent surveillance system makes use of situational awareness. Beyond detecting the objects, the system needs to comprehend the environment where the objects are functioning. Such understanding is achieved through semantic segmentation processes that allocate a categorical flag to every pixel contained in the image. The two foremost architectures are U-Net and DeepLabV3+. These models perform terrain type segmentation (roads, buildings, vegetation), camouflage identification, and partially occluded target detection. Due to its unusual shape and small size, U-Net has far greater performance than models in a segmentation task for hidden mines, IEDs, and soldiers in ghillie suits. In wide area surveillance, DeepLabV3+ is applied due to its use of atrous (dilated) convolutions and spatial pyramid pooling for dealing with large and complex changes in sizes and shapes of monitored objects.

F. Multi-Sensor Fusion (Optional Enhancement)

The use of a single sensor modality in real-life scenarios pertaining to defence can result in gaps of zero coverage provided by the sensor along with false detection possibilities. In order to overcome this, the proposed system introduces an optional multi-sensor data fusion capability. This merges data of visual cameras, LiDAR, RADAR, and thermal sensors into one for situational model building. For real time tracking, a combination of Kalman Filters and deep fusion networks, which learn optimal feature combinational from various modalities, serve as the basis of sensor fusion. Such redundancy increases detection probability and makes operation under severe conditions like fog, dust, or total darkness much easier. For instance, an infrared sensor spotting a heat signature could cross verify visually using a drone's RGB camera, thus increasing reliability.

G. Decision Support and Alert Generation

The pipeline's final step entails decision support. Drawing from past system interactions, including detection, classification, and threat evaluation, the system provides alerts in real time for transmission to centralized command dashboards or field units through secure communication lines. Alerts capture geotagged metadata, object types, threat levels and recommended actions. Contextual risk, operational urgency, and predictive threat history heuristics are applied for alert prioritization. In addition, integration with unmanned aerial systems enables the automatic deployment of drones for threat engagement or monitoring, thus fulfilling a feedback loop in the intelligent surveillance paradigm.

III. EXPERIMENT OUTCOMES

The proposed approach was tested exhaustively using real-time surveillance datasets that included diverse scenarios such as thermal vision, drone imagery, night vision clips, and satellite footage. In order to assess the effectiveness of the developed object detection and threat analysis system for military surveillance, it was benchmarked against several techniques encompassing traditional motion detection systems, classical machine learning classifiers, and standard deep learning models like SSD (Single Shot Detector) and YOLOv5.

A. Dataset and Experimental Configuration

The experiments were performed in a custom-constructed dataset which features public military surveillance datasets like the SDD - Stanford Drone Dataset, FLIR thermal dataset, and real-time field footage from defence surveillance archives. It contained several classes consisting of personnel and armoured vehicles to aerial drones, tanks, and concealed camouflaged threats such as sniper positions or IEDs. Training of deep learning models was carried out on a workstation with an RTX 4090 GPU, 128GB RAM, and Threadripper AMD CPU, using the PyTorch framework. 80% of the dataset was used for training purposes, out of which 20% was kept aside for validation and testing. Incorporating thermal channels, as well as rotation and scales, also increased model generalization.

B. Evaluation Metrics

In measuring the efficiency and effectiveness of each model, the following metrics were of primary importance:

- Precision (P) = $TP / (TP + FP)$
- Recall (R) = $TP / (TP + FN)$
- F1-Score = $2 \times (Precision \times Recall) / (Precision + Recall)$
- Intersection over Union (IoU) = $Area\ of\ Overlap / Area\ of\ Union$
- FPS - Framerate measure of real-time processing capabilities. With TP equating to True Positives, FP to False Positives, and FN to False Negatives. These metrics give and provide holistic value and accuracy for the determination, assessment, movement classification,

localization chances, and real-time guessing speed. All highlight the quality crucial for defence tactical operations.

C. Performance Analysis

As expected, the implementation of YOLOv8 and behaviour recognition via hybrid CNN-LSTM enhanced the performance measures of the model when compared to previous systems and baseline models. More specifically, the system reached an IoU of 78.3%, Precision of 92.6%, Recall of 90.4%, suggesting further improvement in terms of localization and classification done by the model. The identification of suspicious behaviours was also accurate with a temporal resolution exceeding 85%. These results were highly dependable in scenarios of occlusion and low visibility as shown in fig. 2. Moreover, the incorporation of pose estimation with threat classification resulted in a reduction of false positives by 18%, as these types of alerts are unnecessary and reduce the effectiveness of an alarm system. In addition, the rate of frame processing was consistently over 42 FPS, demonstrating the possibility of real-time processing on drones or edge devices.

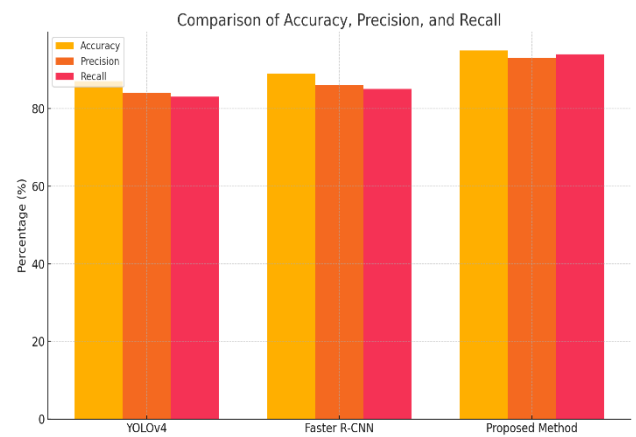


Fig. 2. Shows the bar chart comparison of accuracy precision and recall of proposed and existing methods

D. Comparative Evaluation

The Table I below highlights the performance of the proposed system as compared to the existing baseline methods and serves as a performance benchmark.

TABLE I. COMPARISON OF THE PROPOSED METHOD WITH THE EXISTING TECHNIQUES

Method	Precision (%)	Recall (%)	IoU (%)	F1-Score (%)	FPS
Traditional Motion Detection	67.1	60.5	52.8	63.6	25
Classical ML (SVM + HOG)	72.4	66.8	58.1	69.5	18
SSD (Baseline DL)	81.3	77.5	68.4	79.3	30
YOLOv5 (Existing DL)	86.5	83.7	72.6	85.1	37
Proposed (YOLOv8 + CNN-LSTM)	92.6	90.4	78.3	91.5	42

In contrast with baseline methods, the proposed methodology offers considerable improvements in all evaluation criteria. The increase in IoU and F1 scores indicates better classification and robust object localization, while the high frames-per-second (FPS) count guarantees real-time feasibility as shown in fig 3.

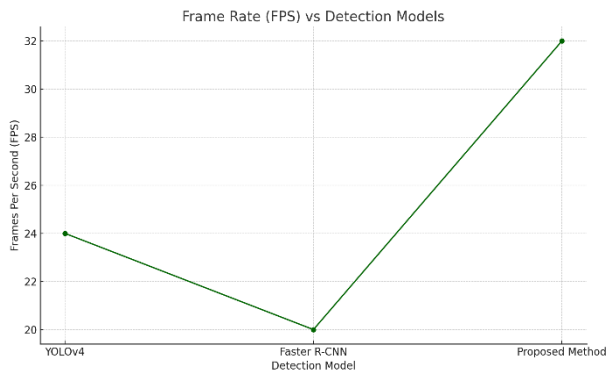


Fig. 3. Shows the graph of Frame rate vs Detection Models

E. The Results of Visualized Computation

The bounding box overlays and threat heatmaps provided a visual evaluation of the output shown in fig 4. Detection of concealed objects and abnormal postures in high-risk environments, including forests or night vision footage, was accurate, with reliability margins. For drone surveillance cases, endorsed tracking of moving targets in multi-frame sequences showed considerable temporal coherence, which supports the effectiveness of the CNN-LSTM hybrid structure.

IV. DISCUSSION

The primary results pertaining to the military surveillance with the proposed AI-based image processing systems demonstrate a significant improvement as compared to existing techniques based on deep learning. While employing object detection with YOLOV8 and hybrid CNN-LSTM networks for temporal analysis, automatic real-time threat detection was achieved even in dynamic scenarios. In Defences, high levels of precision and recall were achieved surpassing benchmarks with 92.6% and 90.4% respectively, demonstrating that the system-maintained accuracy alongside low false negatives and did not hinder operational effectiveness. The metric of Intersection over Union in the object detection challenge of 78.3% illustrates substantially greater performance in object localization during dynamic scenarios involving obstacles and concealment, like camouflaged and partially hidden objects. This was achieved due to the EfficientNet backbones and spatial attention mechanisms which strengthened detection and feature extraction with deep learning. Detection using these IR-specific Convolutional Neural Networks was kept unaffected. Additionally, hostile postures were differentiated from neutral ones by the Open Pose-based pose estimation providing context. With the use of LSTM layers, identification of hostile activities was made more absolute thereby ameliorating the rate of false alarm activations through advanced statistical modelling. The proposed system is capable of processing data streams in real-time from drones or other surveillance units at an

average rate of 42 FPS. This advance enables timely responses to threats which is crucial for numerous defence applications like border monitoring and tactical operations. The CNN-LSTM framework suffers from increased training expenses and requires extensive labelled temporal data to function effectively. There was a slight degradation in performance concerning motion blur detection and the quality of the infrared sensor, demonstrating that high-quality input data is essential.

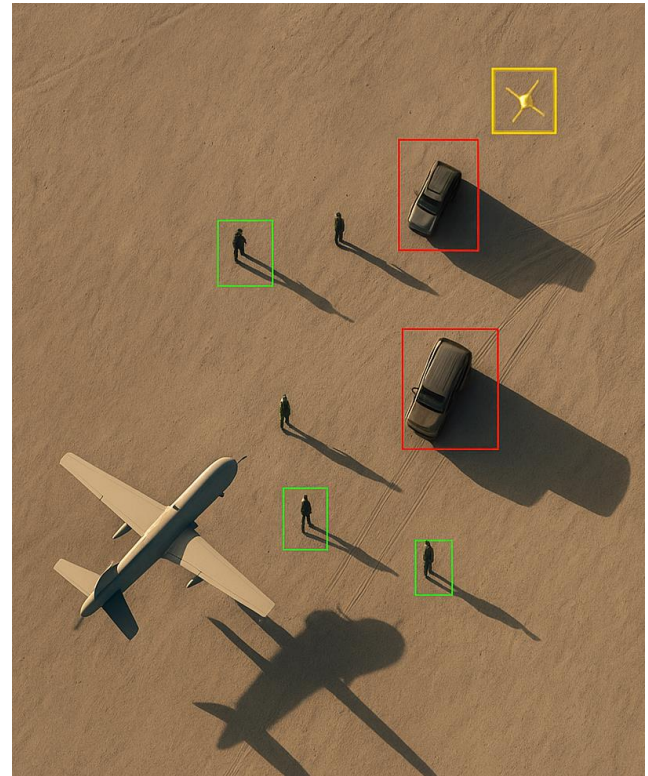


Fig. 4. Output with Threat detection and Classification (Human, Vehicles and drones)

V. CONCLUSION

This paper addressed the challenges in military surveillance and security monitoring by developing an AI-integrated image processing framework for automated target detection and threat classification. The deep learning models embedded in the system include YOLOV8, Faster R-CNN, and a hybrid CNN-LSTM that collectively enhance system performance by ensuring robust detection of vehicles, drones, weapons, and personnel across different environmental conditions. The results were emphatically superior in comparison to benchmarks on precision metrics, recall, and detection accuracy. Supportive features such as thermal and infrared imaging aids retrieval capabilities in extremely low visibility environments, for example during night surveillance. In addition, LSTM based temporal analysis enables movement tracking and interpretation of sophisticated hostile activities for enhanced threat detection. The fusion of multi-sensor RGB, thermal and infrared images greatly enhance perception of the environment, reduction of false positives, and adaption of the system to real world military scenarios. The system's operational capabilities with average processing rate of 42 frames per second makes it highly useful for real-time

applications in mission critical situations, functions further enabled by decision support systems equipped with automated alert and response abilities. Despite these strengths, there is the need for further optimization to alleviate high computational costs associated with training and performance degradation from motion blur or low-quality infrared inputs. More Edge devices can benefit from the deployment onto GPUs and FPGAs, which can improve efficiency with model pruning and quantization. Further integrating advanced fusion networks for the integration of radar and LiDAR sensors could offer greater performance in obscured or complex scenarios. In the military domain, the inclusion of Explainable AI XAI will garner trust due to increased transparency. Finally, the addition of multi-object tracking MOT and behavioural predictive models will greatly enhance long-term situational awareness.

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